

# 26021 Kemi (Polyteknisk grundlag)/26021 Chemistry (Polytechnic Foundation) non- random version - Spring 2025

Der anvendes en scoringsalgoritme, som er baseret på "One best answer"

Dette betyder følgende:

Der er altid netop ét svar som er mere rigtigt end de andre

Studerende kan kun vælge ét svar per spørgsmål

Hvert rigtigt svar giver 1 point

Hvert forkert svar giver 0 point (der benyttes IKKE negative point)

The following approach to scoring responses is implemented and is based on "One best answer"

There is always only one correct answer – a response that is more correct than the rest

Students are only able to select one answer per question

Every correct answer corresponds to 1 point

Every incorrect answer corresponds to 0 points (incorrect answers do not result in subtraction of points)

DK: Hvad er den omtrentlige forventede H-N-H bindingsvinkel i  $N_2H_4$  (hydrazin)?

EN: What is the approximate expected H-N-H bond angle in  $N_2H_4$  (hydrazine)?

Choose one answer

☒ 107°

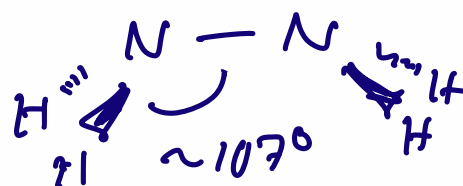
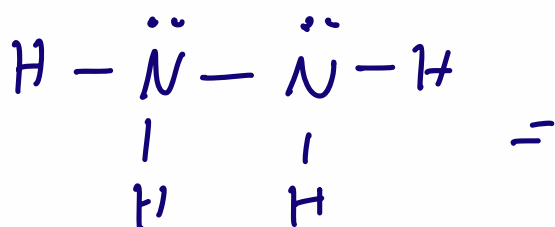
☐ 90°

☐ 120°

☐ 180°

☐ 54°

$$2.5 + 4 \cdot 1 = 14 e^-$$



Smaller than  
tetrahedral angle  
of  $\sim 109.5^\circ$

DK: Hvor mange mol ioner dannes ved komplet opløsning af 1 mol aluminiumsulfat i vand?

EN: How many moles of ions are formed by complete dissolution of 1 mol of aluminum sulfate in water?

Choose one answer

☒ 5 mol

☐ 4 mol

☐ 2 mol

☐ 6 mol



Kapitel 2, p. 92 (table 2.5)

DK: Hvilket af følgende molekyler har ikke en plan geometri?

EN: Which of the following molecules does not possess a planar geometry?

Choose one answer

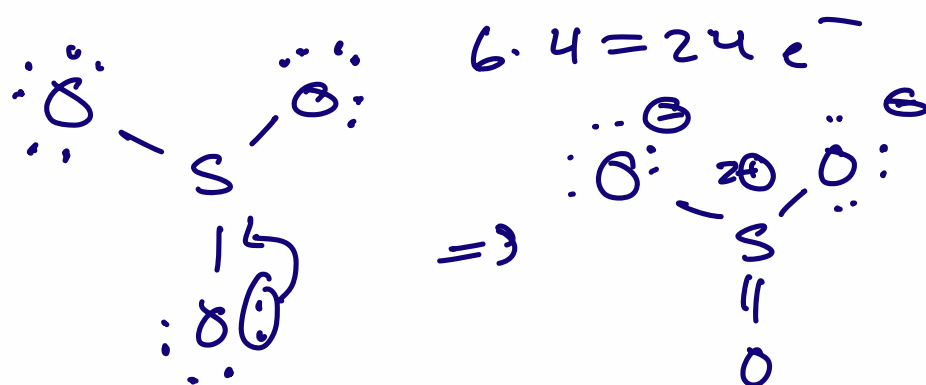
☐  $\text{SO}_3$

☒  $\text{SiH}_4$

☐  $\text{BH}_3$

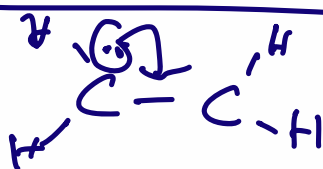
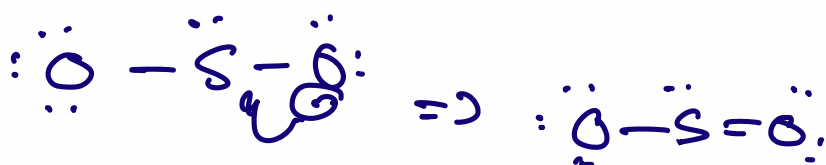
☐  $\text{SO}_2$

☐  $\text{C}_2\text{H}_4$



$\text{BH}_3 \dots$

$18 e^-$



$2 \cdot 4 + 4 = 12 e^-$

DK: Hvilket udsagn er korrekt om en katalysators påvirkning på en kemisk reaktion?

EN: Which statement is correct for a catalyst's influence on a chemical reaction?

Choose one answer

☒ x

Den sænker aktiveringsenergien. *It lowers the activation energy.*

☐

Den bliver forbrugt under reaktionen. *It is consumed during the reaction.*

☐

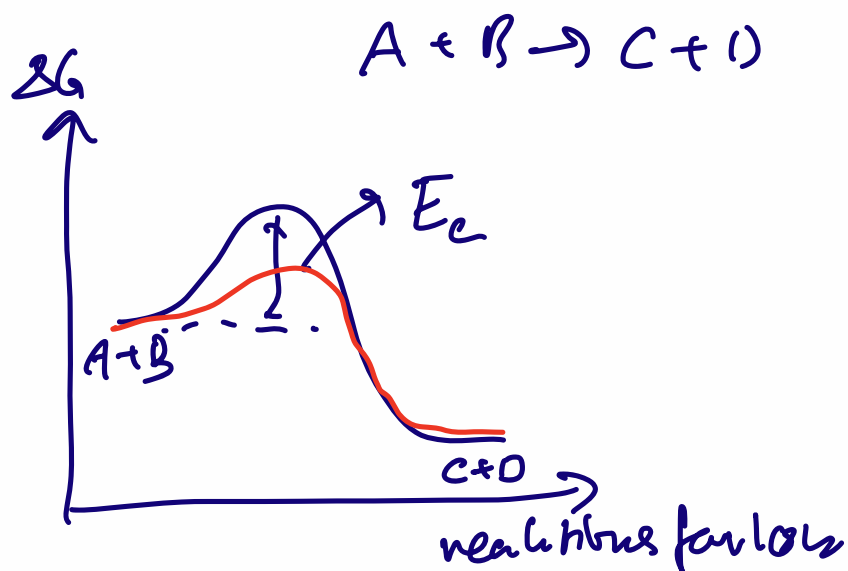
Den hæver aktiveringsenergien. *It raises the activation energy.*

☐

Den forskyder reaktionslignevægten mod produkterne. *It shifts the reaction equilibrium towards the products.*

☐

Den forskyder reaktionslignevægten mod reaktanterne. *It shifts the reaction equilibrium towards the reactants.*



**DK:** Hvad er oxidationstallet for svovl i svovlsyre,  $\text{H}_2\text{SO}_4$ ?

**EN:** What is the oxidation state for sulfur in sulfuric acid,  $\text{H}_2\text{SO}_4$ ?

Choose one answer

☐ +IV (+4)

☒ +VI (+6)

☐ -II (-2)

☐ 0

☐ +II (+2)

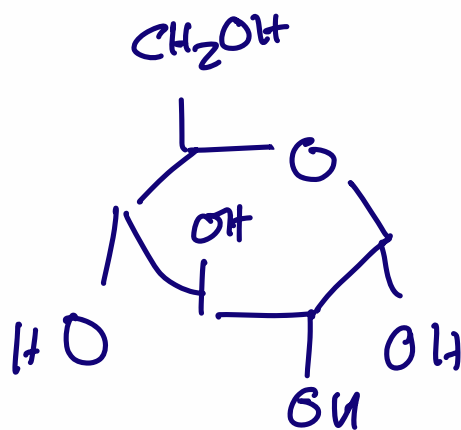
**DK:** Hvilken af følgende forbindelser har det højeste damptryk ved stuetemperatur?

**EN:** Which of the following possess the largest vapor pressure at room temperature?

Choose one answer

- ☐ Vand. *Water.*
- ☐ Glucose. *Glucose.*
- ☐ Natriumchlorid. *Sodium chloride.*

☒ Ethanol. *Ethanol.*



glucose

DK: Hvor mange alkoholer har molekylformlen  $C_4H_{10}O$ ?

EN: How many alcohols have the molecular formula  $C_4H_{10}O$ ?

Choose one answer

☐ 3

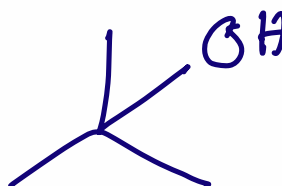
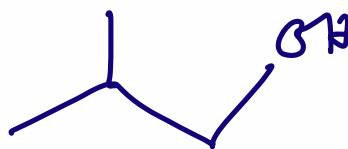
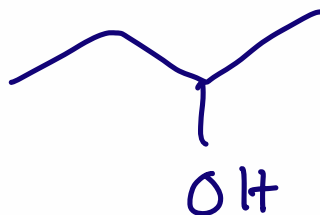
☒ 4

☐ 1

☐ 2

☐ 5

☐ 6





**DK:** Hvor mange elektroner har phosphor i sin yderste skal?

**EN:** How many electrons does phosphorous possess in its outermost shell?

Choose one answer

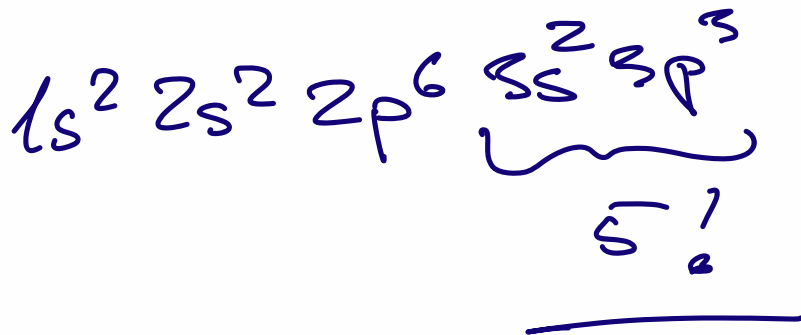
☐ 6

☒ 5

☐ 4

☐ 3

☐ 7

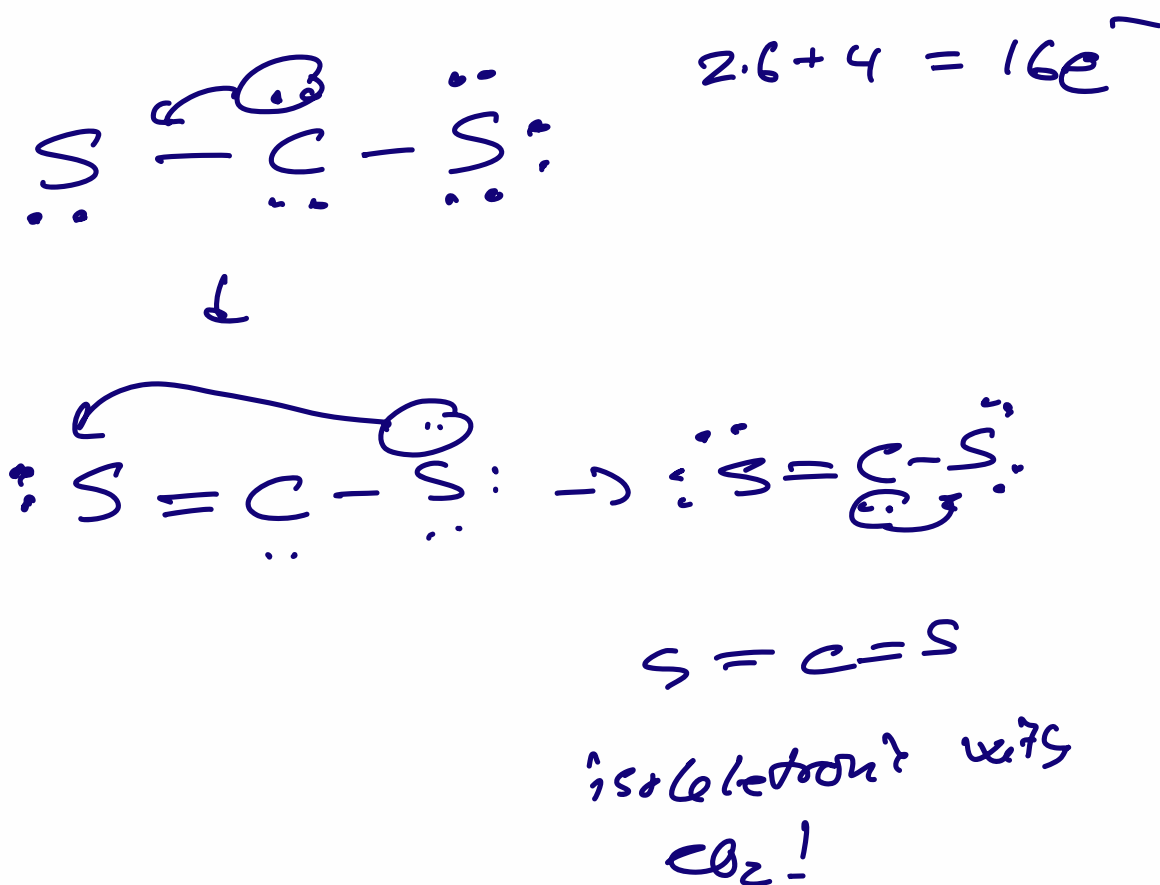


**DK:** Hvilket udsagn er korrekt om CS<sub>2</sub> (som har kulstof som centralatom)?

**EN:** Which statement is correct for CS<sub>2</sub> (which has carbon as a central atom)?

Choose one answer

- ☐ Molekylet er vinklet og polært. *The molecule is bent and polar.*
- ☐ Molekylet er vinklet og upolært. *The molecule is bent and non-polar.*
- ☒ Molekylet er lineært og upolært. *The molecule is linear and non-polar.*
- ☐ Molekylet er lineært og polært. *The molecule is linear and polar.*



DK: Et galvanisk element konstrueres med en halvcelle bestående af  $\text{Fe}^{3+}/\text{Fe}^{2+}$  ( $E^\circ = +0.77 \text{ V}$ ) og en halvcelle bestående af  $\text{Ce}^{4+}/\text{Ce}^{3+}$  ( $E^\circ = +1.6 \text{ V}$ ). Hvilket af følgende udsagn er korrekt?

EN: A galvanic element is constructed with a half-cell consisting of  $\text{Fe}^{3+}/\text{Fe}^{2+}$  ( $E^\circ = +0.77 \text{ V}$ ) and a half-cell consisting of  $\text{Ce}^{4+}/\text{Ce}^{3+}$  ( $E^\circ = +1.6 \text{ V}$ ). Which of the following statements is true?

Choose one answer

- ☐ Elektroner strømmer fra  $\text{Ce}^{3+}$  til  $\text{Fe}^{3+}$ . *Elektrons are flowing from  $\text{Ce}^{3+}$  to  $\text{Fe}^{3+}$*
- ☐  $\text{Fe}^{3+}$  oxideres til  $\text{Fe}^{2+}$ .  *$\text{Fe}^{3+}$  is being oxidized to  $\text{Fe}^{2+}$ .*
- ☐  $\text{Fe}^{2+}$  reduceres til  $\text{Fe}^{3+}$ .  *$\text{Fe}^{2+}$  is being reduced to  $\text{Fe}^{3+}$ .*
- ☒  $\text{Ce}^{4+}$  reduceres til  $\text{Ce}^{3+}$ .  *$\text{Ce}^{4+}$  is being reduced to  $\text{Ce}^{3+}$*
- ☐  $\text{Ce}^{4+}$  oxideres til  $\text{Ce}^{3+}$ .  *$\text{Ce}^{4+}$  is being oxidized to  $\text{Ce}^{3+}$ .*

$$E_{\text{cell}}^\circ = E_{\text{cat}}^\circ - E_{\text{anode}}^\circ > 0$$

$$\Delta G^\circ = -nF E_{\text{cell}}^\circ$$



① has to be the cathode!

**DK:** En given kemisk reaktion har  $\Delta H = -50 \text{ kJ/mol}$  og  $\Delta S = -100 \text{ J/(mol K)}$ . Ved hvilken temperatur skifter reaktionen fra spontan til ikke-spontan?

**EN:** A given chemical reaction has  $\Delta H = -50 \text{ kJ/mol}$  and  $\Delta S = -100 \text{ J/(mol K)}$ . At which temperature does the reaction shift from spontaneous to non-spontaneous?

Choose one answer

☐ 298 K

☐ 600 K

☒ 500 K

☐ 400 K

☐ 273 K

$$\Delta G = \Delta H - T \cdot \Delta S$$

$$\text{Equilibrium: } \Delta G = 0$$

$$\Delta H = T \Delta S$$

$$\Leftrightarrow T = \frac{\Delta H}{\Delta S} = \frac{-50 \cdot 10^3 \frac{\text{J}}{\text{mol}}}{-100 \frac{\text{J}}{\text{K mol}}} = \underline{\underline{500 \text{ K}}}$$

**DK:** En opløsning indeholder både  $\text{Pb}^{2+}$  og  $\text{Ag}^+$ . Ved  $[\text{Cl}^-] = 1.0 \times 10^{-3} \text{ M}$ , hvilket af saltene  $\text{PbCl}_2$  og  $\text{AgCl}$  vil udfælde først, dvs. ved den lavest koncentration af metalionen?

De relevante opløselighedsprodukter er  $K_{\text{sp}}(\text{AgCl}) = 1.8 \times 10^{-10} \text{ M}^2$  og  $K_{\text{sp}}(\text{PbCl}_2) = 1.7 \times 10^{-5} \text{ M}^3$ .

**EN:** A solution contains both  $\text{Pb}^{2+}$  and  $\text{Ag}^+$ . At  $[\text{Cl}^-] = 1.0 \times 10^{-3} \text{ M}$ , which of the salts,  $\text{PbCl}_2$  and  $\text{AgCl}$ , will start to precipitate first, i.e. at the lowest concentration of the metal ion?

The relevant solubility products are  $K_{\text{sp}}(\text{AgCl}) = 1.8 \times 10^{-10} \text{ M}^2$  and  $K_{\text{sp}}(\text{PbCl}_2) = 1.7 \times 10^{-5} \text{ M}^3$ .

Choose one answer

☐ Begge samtidig. Both at the same time.

☐ Kan ikke afgøres. Cannot be determined.

☒  $\text{AgCl}$

☐  $\text{PbCl}_2$

Koncentrationen ved hvilke udfældning vil ske:

$$[\text{Ag}^+] = \frac{K_{\text{sp}}}{[\text{Cl}^-]} = \frac{1.8 \cdot 10^{-10} \text{ M}^2}{1.0 \cdot 10^{-3} \text{ M}} = \underline{\underline{1.8 \cdot 10^{-7} \text{ M}}}$$

$$[\text{Pb}^{2+}] = \frac{K_{\text{sp}}}{[\text{Cl}^-]^2} = \frac{1.7 \cdot 10^{-5} \text{ M}^3}{(1.0 \cdot 10^{-3} \text{ M})^2} = \underline{\underline{17 \text{ M} !}}$$

**DK:** Hvad er den formelle ladning på nitrogenatomet i nitrat ( $\text{NO}_3^-$ ) bestemt ud fra dens Lewis-struktur?

**EN:** What is the formal charge on the nitrogen atom in nitrate ( $\text{NO}_3^-$ ) as determined from its Lewis structure?

Choose one answer

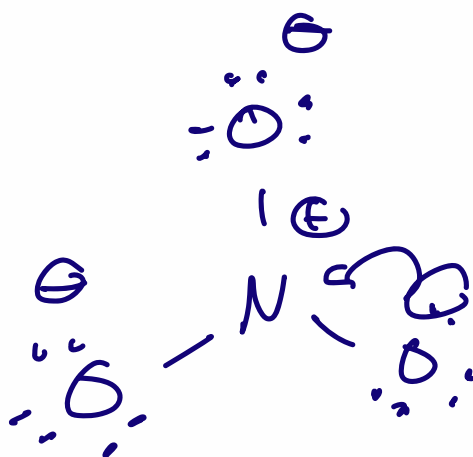
☐ +2

☐ 0

☒ -1

☐ -1

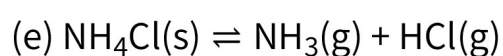
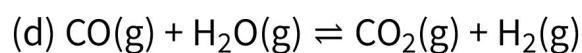
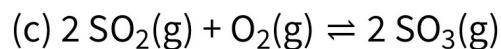
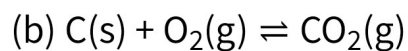
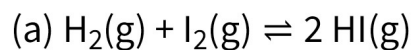
☐ -2



$$5 + 3 \cdot 6 + 1 = 24 e^-$$

**DK:** For hvilken af følgende reaktioner vil en stigning i trykket favorisere dannelsen af produkterne? (se reaktioner under engelsk tekst).

**EN:** For which of the following reactions would an increase in the pressure favor the formation of the products.



Choose one answer

☐ (b)

☐ (d)

☒ (c)

☐ (a)

☐ (e)

for (c)  $K = \frac{p(\text{SO}_3)^2}{p(\text{SO}_2)^2 p(\text{O}_2)}$

$$Q = \frac{4 p(\text{SO}_3)_{\text{start}}}{4 p(\text{SO}_2)_{\text{start}}^2 \cdot 2 p(\text{O}_2)_{\text{start}}} = \frac{1}{2} \frac{p(\text{SO}_3)_{\text{start}}^2}{p(\text{SO}_2)_{\text{start}}^2 p(\text{O}_2)_{\text{start}}}$$

$$Q < K \Rightarrow$$

**DK:** Hvilken molekylær egenskab forklarer det relativt høje kogepunkt for HF i forhold til HCl, HBr og HI?

**EN:** Which molecular property explains the relatively high boiling point for HF in comparison to HCl, HBr, and HI?

Choose one answer

- ☐ Den molare masse. *The molar mass.*
- ☐ Kovalent binding. *Covalent bonding*
- ☐ Ion-dipol-vekselvirkninger. *Ion-dipole interactions*
- ☐ Ion-binding. *Ionic bonding*
- ☒ Hydrogenbinding. *Hydrogen bonding*
- ☐ van der Waals-vekselvirkninger. *van der Waals interactions.*



**DK:** Hvad er pH i en 0.10 M vandig opløsning af eddikesyre ( $\text{CH}_3\text{COOH}$ ,  $\text{p}K_a = 4.7$ ).

**EN:** What is the pH in a 0.1M aqueous solution of acetic acid ( $\text{CH}_3\text{COOH}$ ,  $\text{p}K_a = 4.7$ ).

Choose one answer

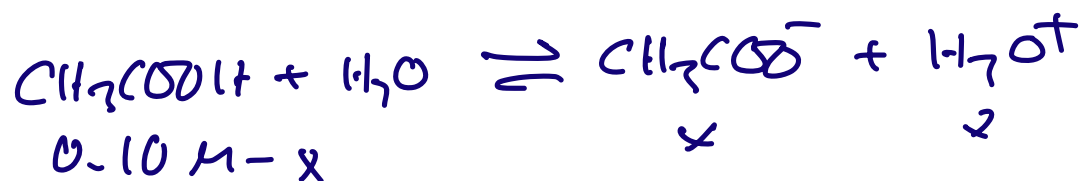
☐ 7.0

☐ 4.7

☒ 2.9

☐ 5.2

☐ 1.0



$$\frac{x^2}{0.10\text{ M} - x} = 10^{-4.7} \text{ M}$$

$$x = 1.403 \cdot 10^{-3} \text{ M}, \quad \text{pH} = \underline{\underline{2.9}}$$

**DK:** En buffer-opløsning blev fremstillet ved at blande 1.00 mol  $\text{NH}_3$  ( $K_b = 1.80 \times 10^{-5} \text{ M}$ ) med 1.00 mol  $\text{NH}_4\text{Cl}$  i vand med et endeligt totalvolumen på 1.00 L. Til 500 mL af denne opløsning blev der tilføjet 30.0 mL af en 1.00 M  $\text{NaOH(aq)}$ -opløsning. Hvad blev pH af denne opløsning?

**EN:** A buffer solution was prepared by mixing 1.00 mol  $\text{NH}_3$  ( $K_b = 1.80 \times 10^{-5} \text{ M}$ ) with 1.00 mol  $\text{NH}_4\text{Cl}$  in water with a final total volume of 1.00 L. To 500 mL of this solution was added 30.0 mL of a 1.00 M  $\text{NaOH(aq)}$  solution. What is the resulting pH of this solution?

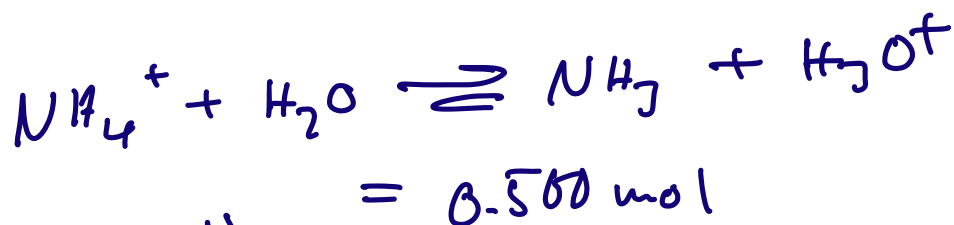
Choose one answer

☒ 9.31

☐ 8.96

☐ 9.83

☐ 9.57



$$n(\text{NH}_4^+)_{\text{start}}$$

$$n(\text{NH}_3)_{\text{start}} = 0.500 \text{ mol}$$

$$n(\text{OH}^-) = 0.0300 \text{ L} \cdot 1.00 = 0.0300 \text{ mol}$$

$$n(\text{NH}_4^+) = 0.500 - 0.0300 = 0.470 \text{ mol}$$

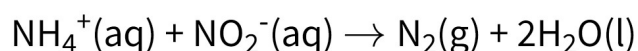
$$n(\text{NH}_3) = 0.500 + 0.0300 = 0.530 \text{ mol}$$

$$K_b = 1.80 \cdot 10^{-5} \text{ M} \Rightarrow pK_b = 4.74$$

$$pK_a = 14 - 4.74 = \underline{9.26}$$

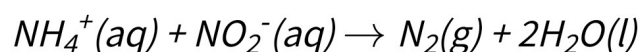
$$pH = pK_a + \log \frac{0.530}{0.470} = \underline{\underline{9.31}}$$

DK: Hvad er hastighedsudtrykket,  $k[\text{NH}_4^+]^x[\text{NO}_2^-]^y$  (hvor  $k$  er en konstant) for reaktionen:



hvor følgende data er blevet bestemt (under engelsk tekst):

EN: What is the rate law,  $k[\text{NH}_4^+]^x[\text{NO}_2^-]^y$  (where  $k$  is a constant) for the reaction:



where the following data have been determined:

| Experiment number | $[\text{NH}_4^+]$ (M) | $[\text{NO}_2^-]$ (M) | Reaction rate (M/s) |
|-------------------|-----------------------|-----------------------|---------------------|
| 1                 | 0.010                 | 0.020                 | 0.020               |
| 2                 | 0.015                 | 0.020                 | 0.030               |
| 3                 | 0.010                 | 0.010                 | 0.005               |

Choose one answer

- ☒  $k[\text{NH}_4^+][\text{NO}_2^-]^2$
- ☐  $k[\text{NH}_4^+]^2[\text{NO}_2^-]$
- ☐  $k[\text{NH}_4^+]^2[\text{NO}_2^-]^2$
- ☐ Ingen af disse
- ☐  $k[\text{NH}_4^+][\text{NO}_2^-]$

Compare Exp #1 and #2:

$$\frac{r_2}{r_1} = \frac{0.030 \text{ M s}^{-1}}{0.020 \text{ M s}^{-1}} = 1.5$$

Same factor for concentration:

$$\frac{0.015 \text{ M}}{0.010 \text{ M}} = 1.5$$

Therefore,  $x = 1$

Compare Exp #1 and #3:

$$\frac{r_1}{r_3} = \frac{0.020 \text{ M s}^{-1}}{0.005 \text{ M s}^{-1}} = 4$$

$$\frac{[\text{NO}_2^-]_1}{[\text{NO}_2^-]_3} = 2, \text{ since } 4 = (2)^2 \text{ we get } y = \underline{\underline{2}}$$

**DK:** Hvilken af følgende ioner har den mindste ionradius?

**EN:** Which of the following ions have the smallest ionic radius?

Choose one answer

☐  $\text{Rh}^+$

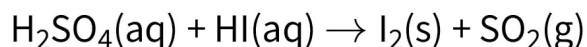
☒  $\text{Sr}^{2+}$

☐  $\text{Br}^-$

☐  $\text{Se}^{2-}$

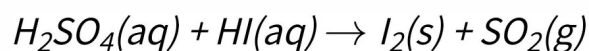
☐ De har alle samme størrelse da de har samme antal elektroner.

**DK:** Afstem følgende reaktion for oxidationen af HI(aq) med svovlsyre.



Hvad er summen af koefficienterne? Brug de laveste mulige, heltallige koefficienter. Glem ikke koefficienter af en (1).

**EN:** Balance the following reaction for the oxidation of HI(aq) with sulfuric acid:



What is the sum of the coefficients? Use the smallest possible, integer coefficients. Do not forget coefficients of one (1).

Choose one answer

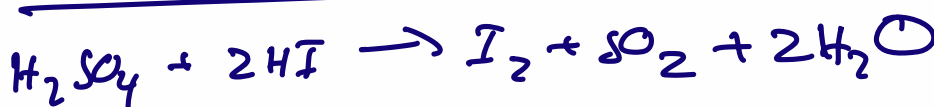
☒ 7

☐ 9

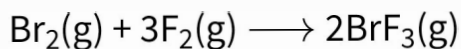
☐ 5

☐ 11

☐ 13

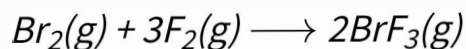


**DK:** Estimer  $\Delta H$  for følgende reaktion



givet bindingsenthalpierne for Br-Br (-192 kJ/mol), F-F (-158 kJ/mol) og Br-F (-197 kJ/mol).

**EN:** Estimate  $\Delta H$  for the following reaction



given the bond enthalpies for Br-Br (-192 kJ/mol), F-F (-158 kJ/mol) and Br-F (-197 kJ/mol).

Choose one answer

☐ -611 kJ/mol

☐ -410 kJ/mol

☒ -516 kJ/mol

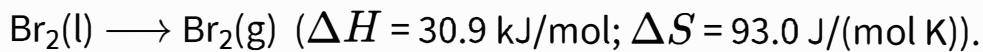
☐ -665 kJ/mol

☐ -720 kJ/mol

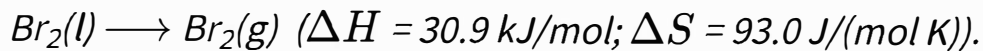
$$\Delta H = \sum H_{\text{bond}}(\text{products}) - \sum H_{\text{bond}}(\text{reactants})$$

$$\begin{aligned} &= 6 \cdot (-197) - (192 + 3 \cdot 158) \\ &= -516 \frac{\text{kJ}}{\text{mol}} \end{aligned}$$

DK: Estimer kogepunktet for Br<sub>2</sub>.



EN: Estimate the boiling point for Br<sub>2</sub>.



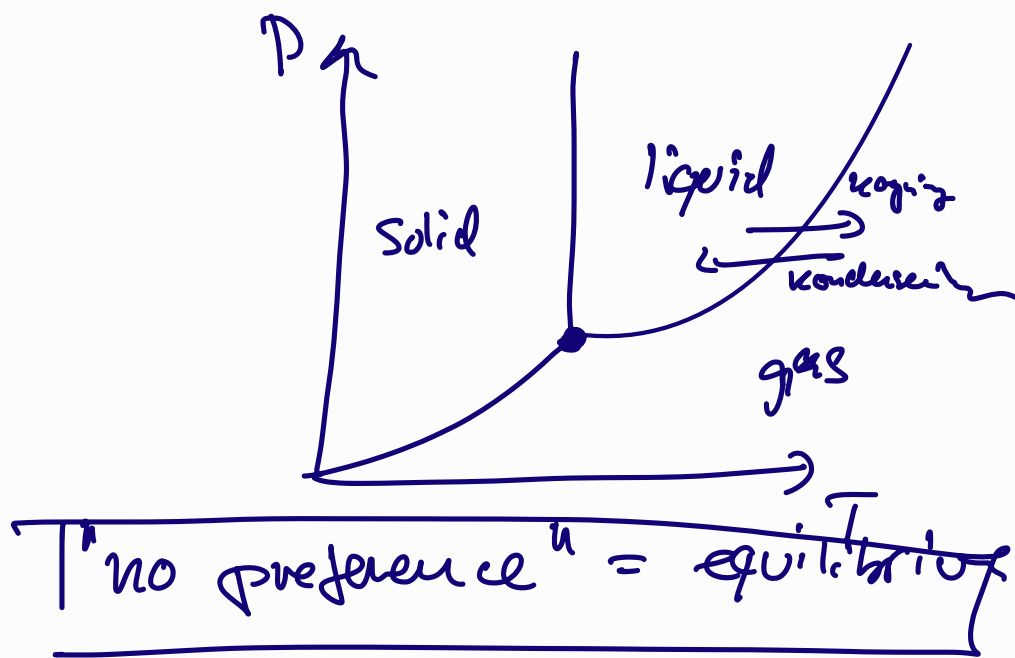
Choose one answer

☒ 59°C

☐ 85°C

☐ 44°C

☐ 373°C



$$\Delta G = \Delta H - T \Delta S$$

$$T = \frac{\Delta H}{\Delta S} = \frac{30.9 \cdot 10^3 \frac{\text{J}}{\text{mol}}}{93.0 \frac{\text{J}}{\text{mol K}}} = 332 \text{ K}$$

$$332 - 273 = \underline{\underline{59^\circ \text{C}}}$$

DK: Hvilket af følgende udsagn er korrekt for en 0.10 M opløsning af en svag syre, HX?

EN: Which of the following statements is correct for a 0.10 M solution of a weak acid, HX?

Choose one answer

☒  $[HX] > [H^+]$

☐ pH = 1

☐ pH = 1,  $[H^+] = 0.10 \text{ M}$

☐  $[H^+] = 0.10 \text{ M}$

☐  $[X^-] = 0.10 \text{ M}$

✓  
 $\rightarrow [H_3O^+] = 0.1 \text{ M}$  f 16

$\rightarrow$  same

$\rightarrow$  same

$\rightarrow$  same



**DK:** Hvad er den molære opløselighed af  $\text{Cu}(\text{OH})_2$  i vand givet at  $K_{\text{sp}}(\text{Cu}(\text{OH})_2) = 1.6 \times 10^{-19} \text{ M}^3$ ?

**EN:** What is the molar solubility of  $\text{Cu}(\text{OH})_2$  in water given that  $K_{\text{sp}}(\text{Cu}(\text{OH})_2) = 1.6 \times 10^{-19} \text{ M}^3$ ?

Choose one answer

☐  $6.3 \times 10^{-10} \text{ M}$

☒  $3.4 \times 10^{-7} \text{ M}$

☐  $2.0 \times 10^{-5} \text{ M}$

☐  $1.6 \times 10^{-19} \text{ M}$

☐  $2.4 \times 10^{-20} \text{ M}$

$$x \cdot (2x)^2 = 1.6 \cdot 10^{-19} \text{ M}^3$$
$$x = \underline{\underline{3.4 \cdot 10^{-7} \text{ M}}}$$

**DK:** Atomet med elektronkonfigurationen  $4s^24p^5$  i yderste skal vil være i:

**EN:** The atom with the electronic configuration  $4s^24p^5$  in the outermost shell will be in:

Choose one answer

☐ 4. hovedgruppe, 7. periode.  $4^{th}$  main group,  $7^{th}$  period.

☒ 7. hovedgruppe, 4. periode.  $7^{th}$  main group,  $4^{th}$  period.

☐ 7. hovedgruppe, 5. periode.  $7^{th}$  main group,  $5^{th}$  period.

☐ 5. hovedgruppe, 4. periode.  $5^{th}$  main group,  $4^{th}$  period.

☐ 4. hovedgruppe, 5. periode.  $4^{th}$  main group,  $5^{th}$  period.

**DK:** En blanding af 0.50 mol  $H_2(g)$  og 0.50 mol  $N_2(g)$  fyldes i en 25 L-beholder ved  $30^\circ C$ . Beholderen punkteres med en nål. Efter noget tid, hvilket af følgende udsagn er korrekt?

**EN:** A mixture of 0.50 mol  $H_2(g)$  and 0.50 mol  $N_2(g)$  are placed in a 25 L container at  $30^\circ C$ . The container is punctured with a needle. After some time, which of the following statements is correct?

Choose one answer

- ☐ Kun partialtrykket af  $H_2$  er steget. *Only the partial pressure of  $H_2$  has increased.*
- ☐ Partialtrykkene af de to gasser er identisk. *The partial pressures of the gases are identical.*
- ☒ Partialtrykket af  $N_2$  er større end partialtrykket af  $H_2$ . *The partial pressure of  $N_2$  is larger than the partial pressure of  $H_2$ .*
- ☐ Partialtrykket af  $H_2$  er større end partialtrykket af  $N_2$ . *The partial pressure of  $H_2$  is larger than the partial pressure of  $N_2$ .*
- ☐ Partialtrykkene af begge gasser er forøget. *The partial pressures of both gases have increased?*

**DK:** En opløsning blev fremstillet ved at opløse 9.81 g af et ukendt molekyle (ikke-flygtig og ikke-elektrolyt) i 90.0 g vand. Kogepunktet for opløsningen blev bestemt til  $100.37^\circ\text{C}$  ved 1.00 atm. Hvad er den omtrentlige molare masse af det ukendte molekyle? (For vand er  $K_b = 0.51^\circ\text{C}/m$ ).

**EN:** As solution was prepared by dissolving 9.81 g of an unknown molecule (non-volatile and non-electrolyte) in 90.0 g of water. The boiling point of the solution was determined to be  $100.37^\circ\text{C}$  at 1.00 atm. what is the approximate molar mass of the unknown molecule? ( $K_b = 0.51^\circ\text{C}/m$  for water).

Choose one answer

☐ 480 g/mol

☒ 150 g/mol

☐ 300 g/mol

☐ 52 g/mol

☐ 18 g/mol

$$\Delta T_B = 0.37^\circ\text{C}$$

$$\boxed{\Delta T_B = i K_B m}$$

$$\text{Since } i = 1;$$

$$m = \frac{\Delta T_B}{K_B} = \frac{0.37^\circ\text{C}}{0.51^\circ\text{C}/m} \rightarrow \text{mol/kg}$$

$$= 0.725 \frac{\text{mol}}{\text{kg}}$$

$$n = 0.725 \frac{\text{mol}}{\text{kg}} \cdot 0.0900 \text{ kg}$$

$$= 0.0653 \text{ mol}$$

$$M = \frac{9.81 \text{ g}}{0.0653 \text{ mol}} = \underline{\underline{150 \text{ g/mol}}}$$

DK: Betragt følgende:  $^{72}\text{Zn}$ ,  $^{75}\text{As}$ ,  $^{74}\text{Ge}$ . Disse atomer har:

EN: Consider the following:  $^{72}\text{Zn}$ ,  $^{75}\text{As}$ ,  $^{74}\text{Ge}$ . These atoms have:

Choose one answer



☒ Det samme antal neutroner. *The same number of neutrons.*

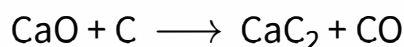
☐ Det samme antal protoner. *The same number of protons.*

☐ Det samme antal elektroner. *The same number of electrons.*

☐ Det samme antal protoner og neutroner. *The same number of protons and neutrons.*

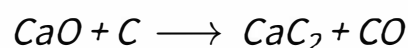
☐ Det samme nukleontal (mass number). *The same mass number.*

**DK:** Calciumcarbid,  $\text{CaC}_2$ , er et vigtigt industrielt kemikalie der fremstillet ved at opvarme calciumoxid med kulstof jævnt før følgende, ikke-afstemte reaktionsskema:



Hvor meget  $\text{CaC}_2$  kan der maksimalt produceres fra reaktionen af  $\text{CaO}$  (i overskud) med 10.2 g kulstof (C)?

**EN:** Calcium carbide,  $\text{CaC}_2$ , is an important industrial chemical that is prepared by heating  $\text{CaO}$  with carbon according to the following, non-balanced reaction scheme:



What is the maximum amount of  $\text{CaC}_2$  that can be produced from the reaction of  $\text{CaO}$  (in excess) with 10.2 g carbon (C)?

Choose one answer

☐ 6.0 g

☒ 18.1 g

☐ 38.4 g

☐ 54 g

☐ 27.2 g



$$m(\text{C}) = 10.2 \text{ g}$$

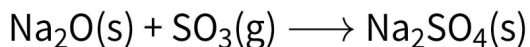
$$n(\text{C}) = \frac{10.2 \text{ g}}{12.0 \text{ g/mol}} = 0.850 \text{ mol}$$

$$n(\text{CaC}_2) = \frac{1}{3} \cdot 0.850 \text{ mol} = 0.283 \text{ mol}$$

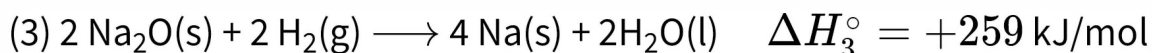
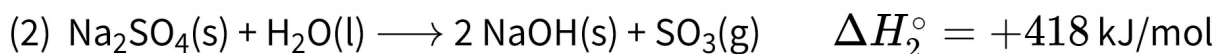
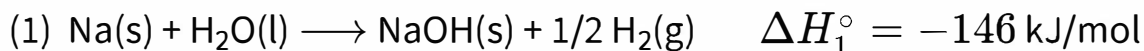
$$M(\text{CaC}_2) = 40.1 + 2 \cdot (12.0) = 64.1 \text{ g/mol}$$

$$m = 0.283 \text{ mol} \cdot 64.1 \text{ g/mol} \\ = \underline{\underline{18.1 \text{ g}}}$$

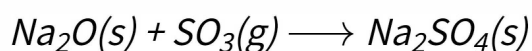
DK: Udregn  $\Delta H^\circ$  for reaktionen:



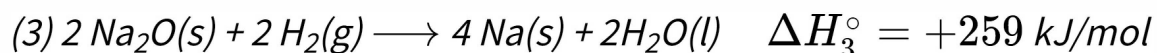
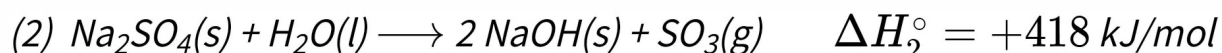
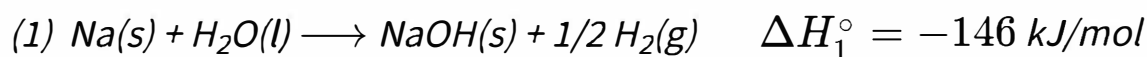
givet følgende information:



EN: Calculate  $\Delta H^\circ$  for the reaction



given the following information:



Choose one answer

☐ -710 kJ/mol

☒ -581 kJ/mol

☐ 581 kJ/mol

☐ 710 kJ/mol

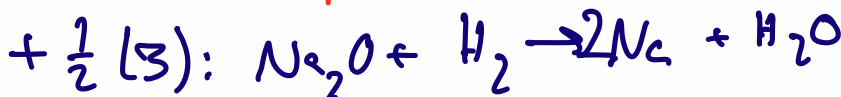
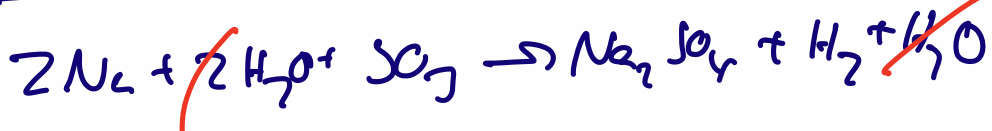
☐ -451 kJ/mol

☐ 451 kJ/mol

☐ +387 kJ/mol

☐ -387 kJ/mol

2 · (1) - 2 ·



$$\Delta H^\circ = 2 \cdot (-146) - 418 + \frac{1}{2} \cdot 259$$

